Case Study 3: Predicting Titanic Survival with Logistic Regression

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Section: CSE-3

Year: III Year, V Semester (2026-27)

Subject: Fundamentals of Data Science

Faculty: <FACULTY NAME>

College: Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad

Date of Submission: 26 October 2026

1. Problem Statement and Objectives

The sinking of the Titanic is one of the best known shipwrecks, and survival was not random: some groups of passengers had far better chances than others. The problem in this case study is to predict, from basic passenger details, whether a passenger survived (Survived = 1) or not (0). This is a binary classification problem, and logistic regression is a natural first model because it gives a survival probability and coefficients that can be interpreted.

The objectives, following the brief, were to:

- explore the data with summary statistics and visualisations and describe the patterns found;
- handle missing values and encode categorical variables without leaking test data;
- build and train a logistic regression model with scikit-learn;
- evaluate it on unseen test data with accuracy, precision, recall, F1-score, ROC-AUC and the ROC curve;
- interpret the coefficients and test which features are statistically significant; and
- deploy the trained model as a Streamlit web app.

2. Dataset Description

The dataset (titanic.csv) was provided by the course faculty. It has **891 rows** (one per passenger) and **12 columns**. The target is **Survived**: 342 passengers survived and 549 did not, an overall survival rate of **38.38%**. The classes are moderately imbalanced, so a model that always predicts 'did not survive' would already be about 61.62% accurate.

Table 1: Features of the Titanic dataset

Feature	Type	Meaning	Used in model
Pclass	Integer (ordinal)	Ticket class: 1 upper, 2 middle, 3 lower	Yes
Sex	Categorical	male / female	Yes
Age	Numeric	Age in years	Yes
SibSp	Integer	Siblings or spouses on board	Yes
Parch	Integer	Parents or children on board	Yes
Fare	Numeric	Ticket fare	Yes
Embarked	Categorical	Port: C Cherbourg, Q Queenstown, S Southampton	Yes
PassengerId, Name, Ticket	ID / text	Identifiers, unique or almost unique	No
Cabin	Text	Cabin number, mostly missing	No
Survived	Binary	Target: 1 survived, 0 did not	Target

Data quality issues. Cabin is missing for 687 passengers (77.10%), Age is missing for 177 passengers (19.87%), Embarked is missing for 2 passengers (0.22%). All other columns are complete. Fare is strongly right skewed (median 14.45, mean 32.20, maximum 512.33).

3. Methodology

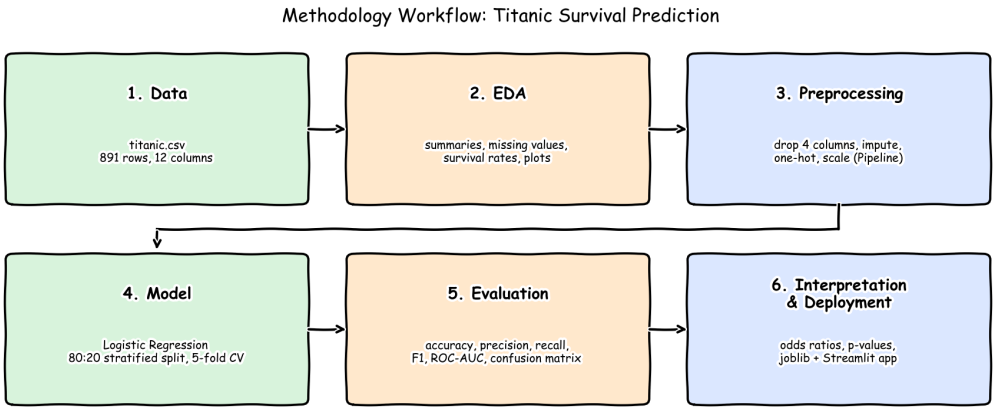


Figure 1: Methodology workflow followed in this case study

Preprocessing

- **Dropped columns:** PassengerId (row number), Name and Ticket (almost unique text) and Cabin (77.10% missing, so imputing it would mostly be guessing).
- **Numeric features** (Pclass, Age, SibSp, Parch, Fare): median imputation (only Age has gaps; the median is robust to skew), then StandardScaler so that every coefficient means the effect of a one standard deviation increase and the coefficients can be compared.
- **Categorical features** (Sex, Embarked): most frequent value imputation (Embarked has 2 gaps), then one-hot encoding with the first level dropped (baselines: female, Cherbourg). Dropping one level avoids perfectly collinear dummies, which is needed for the statsmodels significance test.
- **No leakage:** all steps sit in one scikit-learn Pipeline (ColumnTransformer + model), so medians, modes, means and standard deviations are learned from the training data only.

Train/test split and model

The data was split 80:20 with stratification on Survived and random_state=42, giving 712 training and 179 test passengers with the same survival rate. The model is LogisticRegression(max_iter=1000) with scikit-learn's default L2 penalty (C=1). Stability was checked with 5-fold stratified cross-validation on the training set, refitting the whole pipeline in every fold. For p-values and confidence intervals a statsmodels Logit model was fitted on the same preprocessed training data.

Evaluation metrics and why they were chosen

- **Accuracy**: share of all passengers classified correctly; easy to read but hides which class the errors come from.
- **Precision**: of the passengers predicted to survive, how many really did.
- **Recall**: of the passengers who really survived, how many the model found.
- **F1-score**: harmonic mean of precision and recall, a single score for the survivor class.
- **ROC-AUC**: how well the predicted probabilities rank survivors above non-survivors, independent of the 0.5 threshold.
- **Confusion matrix**: the exact counts of correct and wrong predictions of each kind.

4. Statistical Findings

Table 2: Summary statistics of the numeric features

Statistic	Age	Fare	SibSp	Parch	Pclass
Count	714	891	891	891	891
Mean	29.70	32.20	0.52	0.38	2.31
Std. dev.	14.53	49.69	1.10	0.81	0.84
Min	0.42	0.00	0.00	0.00	1.00
Median	28.00	14.45	0.00	0.00	3.00
Max	80.00	512.33	8.00	6.00	3.00

Age has only 714 known values and ranges from 0.42 to 80 years (median 28). Most passengers travelled without siblings, spouses, parents or children (median 0 for SibSp and Parch), and the Pclass median of 3 shows that at least half of them were in third class.

Table 3: Survival rate by sex, passenger class and port of embarkation

Group	Passengers	Survived	Survival rate
Sex = female	314	233	74.20%
Sex = male	577	109	18.89%
Pclass = 1	216	136	62.96%
Pclass = 2	184	87	47.28%
Pclass = 3	491	119	24.24%
Embarked = C	168	93	55.36%
Embarked = Q	77	30	38.96%
Embarked = S	644	217	33.70%

The differences between groups are large. **74.20% of females survived against only 18.89% of males.** Survival also falls steadily with class: 62.96% in first class, 47.28% in second and 24.24% in third. Passengers who boarded at Cherbourg survived at 55.36% compared with 33.70% at Southampton.

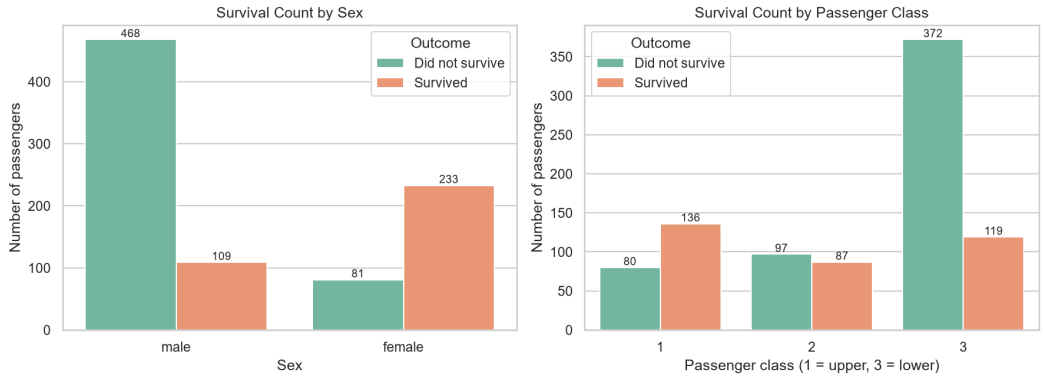


Figure 2: Survival counts by sex and by passenger class

Figure 2 shows the same pattern in counts. Sex and class also combine: **96.8% of first class females survived but only 13.5% of third class males did**, and third class females (50.0%) still did better than first class males (36.9%).

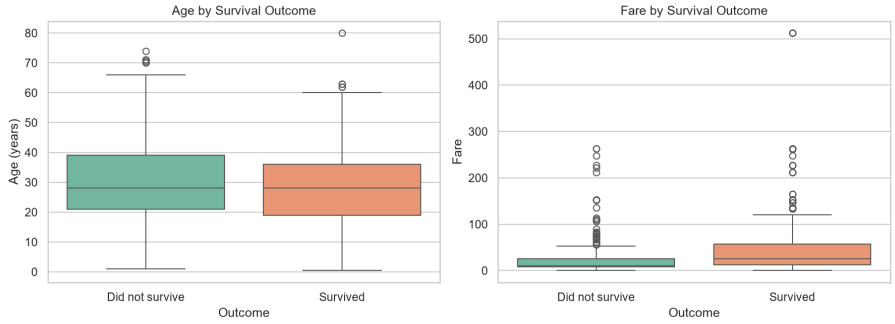


Figure 3: Age and Fare by survival outcome

The median age is 28.0 for survivors and 28.0 for non-survivors, so age alone does not separate the groups. Fare shows a clear shift: survivors paid a median of 26.0 against 10.5, consistent with wealthier passengers having better access to lifeboats.

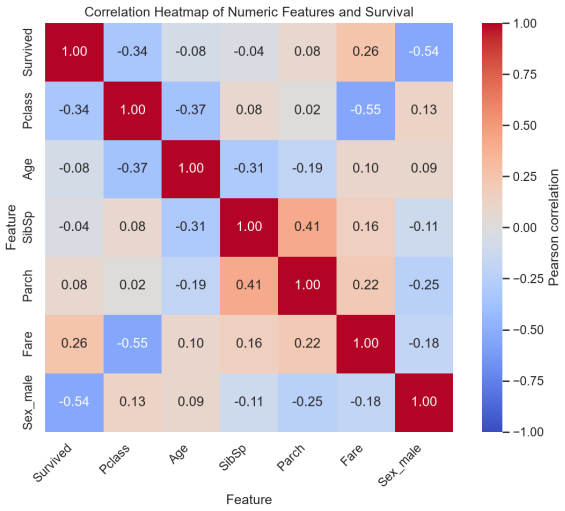


Figure 4: Correlation heatmap of numeric features, Sex (1 = male) and survival

Sex_male has the strongest correlation with survival (-0.543), followed by Pclass (-0.338) and Fare (0.257). Age (-0.077), SibSp (-0.035) and Parch (0.082) are only weakly correlated with it. Pclass and Fare are strongly correlated with each other (-0.549), so they carry overlapping information, which becomes important when testing significance in Section 6.

5. Model Evaluation

Five-fold cross-validation on the training set gave fold accuracies of 0.7972, 0.7972, 0.8028, 0.7465, 0.8310, a mean of **0.7949** with standard deviation 0.0273. The training accuracy was 0.8076, close to the CV mean, so the model is stable and not overfitting. Table 4 gives the results on the 179 unseen test passengers.

Table 4: Test set performance of the logistic regression model

Metric	Value	Meaning here
Accuracy	0.8045	Correct for 144 of 179 passengers (baseline 0.6145)
Precision	0.7931	46 of 58 predicted survivors really survived
Recall	0.6667	46 of 69 real survivors were found
F1-score	0.7244	Balance of precision and recall for survivors
ROC-AUC	0.8435	Ranking quality of the survival probabilities

The model is clearly better than the 0.6145 majority baseline. The classification report shows it is stronger on non-survivors (recall 0.8909, F1 0.8485) than on survivors (recall 0.6667, F1 0.7244), which is expected because non-survivors are the majority class and the default 0.5 threshold favours them.

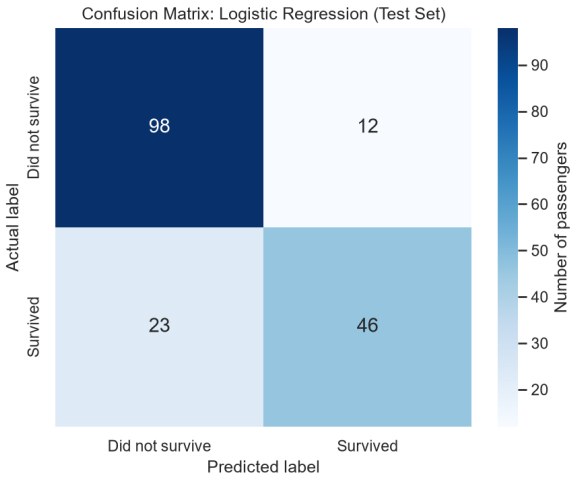


Figure 5: Confusion matrix on the test set

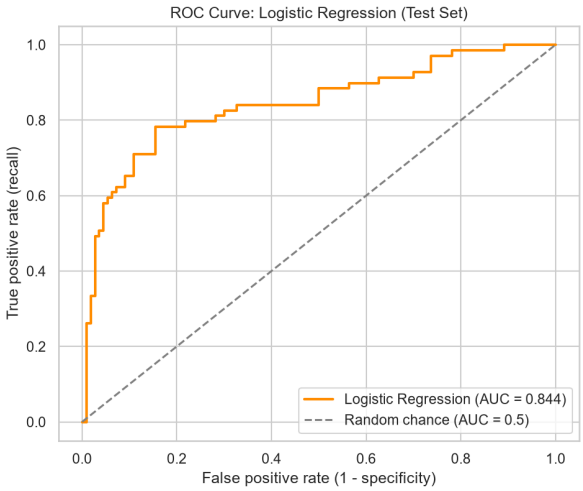


Figure 6: ROC curve with the random-chance diagonal

In Figure 5, of the 179 test passengers, 98 non-survivors and 46 survivors were classified correctly. **The 12 false positives** are passengers the model was too optimistic about (predicted to survive but died). **The 23 false negatives** are real survivors the model wrote off. Missed survivors are the more common error, which is why recall is the weakest metric. If finding survivors mattered more, the threshold could be lowered below 0.5 to raise recall at the cost of precision.

The ROC curve in Figure 6 stays well above the diagonal. An **AUC of 0.8435** means that for a random survivor and a random non-survivor, the model gives the survivor the higher probability about 84% of the time.

6. Interpretation

Coefficients and odds ratios

Numeric features were standardised, so their coefficients are the change in the log-odds of survival for a one standard deviation (SD) increase; for dummy columns they compare with the baseline (female, Cherbourg). An odds ratio below 1 lowers the odds of survival and above 1 raises them.

Table 5: Scikit-learn coefficients and odds ratios (sorted by size) with statsmodels p-values and 95% confidence intervals

Feature	Coefficient	Odds ratio	p-value	95% CI of odds ratio	Significant (0.05)
Sex_male	-2.5598	0.0773	< 0.0001	0.044 to 0.105	Yes
Pclass	-0.9158	0.4002	< 0.0001	0.295 to 0.504	Yes
Age	-0.5009	0.6060	< 0.0001	0.477 to 0.743	Yes
Embarked_S	-0.3835	0.6815	0.1384	0.404 to 1.134	No
Embarked_Q	0.2885	1.3345	0.4706	0.578 to 3.276	No
SibSp	-0.2577	0.7728	0.0367	0.594 to 0.984	Yes
Fare	0.1017	1.1071	0.5058	0.837 to 1.436	No
Parch	-0.0598	0.9419	0.5036	0.758 to 1.146	No

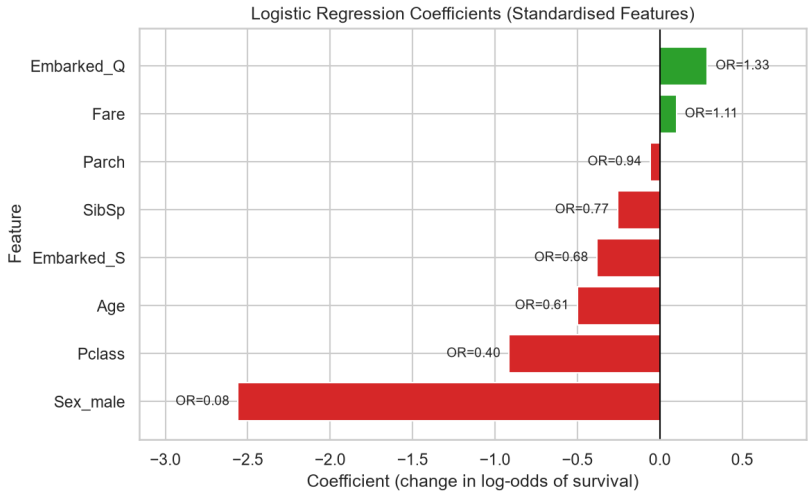


Figure 7: Logistic regression coefficients with odds ratios (OR)

- **Sex_male** (coefficient -2.5598, OR 0.0773): being male multiplies the odds of survival by about 0.08, roughly 92% lower odds than a female with the same other details. This is by far the largest effect.
- **Pclass** (OR 0.4002, 1 SD = 0.83 classes): moving towards third class cuts the odds by about 60% per SD.
- **Age** (OR 0.6060, 1 SD = 13.02 years): older passengers had about 39% lower odds per SD. Age matters once sex and class are controlled, even though the medians in Figure 3 were equal.
- **SibSp** (OR 0.7728, 1 SD = 1.06): more siblings or spouses lowered the odds by about 23% per SD.
- **Embarked, Fare and Parch** have smaller effects (Embarked_S OR 0.68, Embarked_Q OR 1.33, Fare OR 1.11, Parch OR 0.94).

Statistical significance

The statsmodels Logit model converged with a pseudo R-squared of 0.3449 and a likelihood ratio test p-value of 7.44e-66, so the features together explain survival far better than an intercept-only model. At the 0.05 level, **Pclass, Age, SibSp, Sex_male** are significant. SibSp is only just significant (p = 0.0367, upper CI of the odds ratio 0.984). Parch, Fare, Embarked_Q, Embarked_S are **not** significant: their odds ratio confidence intervals include 1. Fare was correlated with survival in the EDA (0.257), but once Pclass is in the model it adds little because the two overlap (correlation -0.549). Likewise, the port differences seen in Table 3 mostly disappear once sex and class are accounted for. The unpenalised statsmodels coefficients are slightly larger than scikit-learn's (Sex_male -2.6902 vs -2.5598) because of the L2 penalty, but all signs and the ranking agree.

Deployment with Streamlit

The complete fitted pipeline (imputers, encoder, scaler and model) was saved with joblib to `3/model/titanic_logreg.joblib`. The Streamlit app `3/app.py` loads it once with `st.cache_resource` (path relative to the app file), offers inputs for passenger class, sex, age, fare, siblings/spouses, parents/children and port of embarkation (shown by full name), and on clicking **Predict** shows the survival probability, a progress bar and the predicted class. Because the whole pipeline is saved, raw inputs are processed exactly as in training. It runs locally with `streamlit run 3/app.py` and can be deployed on Streamlit Community Cloud by selecting the GitHub repository, branch main and main file path 3/app.py.

7. Conclusion and Limitations

Survival on the Titanic was strongly linked to sex and class: 74.20% of females and 62.96% of first class passengers survived against 18.89% of males and 24.24% of third class passengers. A leak-free logistic regression pipeline reached **accuracy 0.8045, precision 0.7931, recall 0.6667, F1 0.7244 and ROC-AUC 0.8435** on unseen data, with a stable cross-validation accuracy of 0.7949. Sex, passenger class, age and number of siblings/spouses were the significant predictors, and the model was deployed as a working Streamlit app.

Limitations:

- The dataset is small (891 passengers, 179 in the test set), so each test error changes the metrics noticeably.
- Age is 19.87% missing and was filled with one median value, which hides real variation; Cabin information was discarded entirely.
- Recall for survivors is only 0.6667 at the default 0.5 threshold; 23 survivors in the test set were missed.
- Logistic regression assumes each feature has a linear effect on the log-odds, so interactions such as sex combined with class are only partly captured.

8. References

- Titanic passenger dataset (titanic.csv, 891 rows), provided by the course faculty for this case study.
- Libraries: pandas 3.1.0, numpy 2.5.4, matplotlib 3.11.2, seaborn 0.13.2, scikit-learn 1.9.1, statsmodels 0.15.0, joblib 1.6.0, streamlit 1.66.1, reportlab 5.1.0.
- Streamlit Community Cloud deployment documentation: https://docs.streamlit.io/streamlit-community-cloud/deploy-your-app
- GitHub repository: <GITHUB REPO URL>

9. AI Assistance Acknowledgement

Code structure drafted with AI assistance (Claude); executed, verified and interpreted by Nikhil Kotte. Dataset sources: titanic.csv (891 passengers) provided by the course faculty.